Aljon A. Novelo

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TECHNICAL SKILLS

Programming: Java, Python, Bash, TypeScript, SQL, JavaScript, C, Dart, R, Assembly, C++
Databases: MongoDB, PostgreSQL, Firebase, Qdrant
Infrastructure: Docker, Google Cloud Platform (GCP), Amazon Web Services (AWS), Linux, Inngest
Testing: Playwright, Mocha, Chai, Postman, Integration, End-to-End
Collaboration: GitLab, GitHub, Redmine, CI/CD, Agile

WORK EXPERIENCE

Web Developer Intern | i-BRIDGE Systems Philippines Inc. | July - August 2025

- Contributed to Agile sprint management and development workflows using GitLab, Redmine, and Microsoft Office for collaboration, task tracking, and project documentation
- Designed and implemented additional user interface components for an internal tool using the MERN stack
- Developed and integrated additional database fields to support the new components
- Conducted preliminary unit and integration testing using Mocha and Chai to resolve issues before
- Deployed the new features to Google Cloud Platform

PROJECTS

Retrieval-Augmented Generation (RAG) using Google Gemini | Artificial Intelligence | Python

- Developed a local Python RAG application for querying PDFs using Google Gemini API
- Utilized Llama Index for chunking PDFs and Qdrant for storing it in a vector database
- Integrated an Inngest development server for logging and monitoring the application

Learning Platform for LaTeX Writers | Web Development | MERN Stack

- Conducted interviews with 20 students to identify requirements for a LaTeX-focused learning platform
- Developed the platform using the MERN stack
- Implemented a recursive-descent parser to process LaTeX input
- Achieved a System Usability Scale score of 83.25 and performed thematic analysis of interview responses to identify areas for improvement

Color-Based Image Segmentation | Digital Image Processing | C++

- Implemented a K-means clustering algorithm in C++ to segment images based on dominant color clusters
- Processed 1,000+ sequential JPEG frames
- Demonstrated the project through a video presentation showcasing unsupervised clustering for image segmentation

Context-Free Grammar Simulator | Compiler Design | Python

- Developed a Python-based command-line simulator that performs LL parsing
- Designed the simulator to support unambiguous grammars
- Implemented file-based input parsing for grammar definitions

Computer Architecture Simulation | Computer Architecture | VHDL

- Designed and simulated combinational and sequential circuits using VHDL, including test benches for validating logic behavior
- Evaluated instruction performance and system efficiency through CPI analysis, speedup computations, and execution time comparisons
- Conducted benchmark testing to analyze real-world CPU performance, using emulator tools and hardware specification assessments

Mobile Health Monitoring System | Mobile Development | Flutter

- Developed a QR scanner for building staffs in recording entries of students and other constituents

Institute Room Reservation System | Web Development | PERN Stack

- Implemented dynamic data tables, notifications, and booking workflows for real-time interaction
- Integrated admin and user-specific features, such as class schedules, announcements, and reports

Networking Game | Data Communications and Networking | Java
- Implemented real-time player interactions using Java sockets for seamless communication
- Utilized multithreading to optimize network performance and ensure responsive gameplay

Programming Language Interpreter | Compiler Design | Python
- Built a parser that constructs an abstract syntax tree to validate and process LOLCODE programs
- Implemented execution logic for core LOLCODE constructs, including variable assignments, conditionals, loops, and arithmetic operations

EDUCATIONAL BACKGROUND

Bachelor of Science in Computer Science | University of the Philippines - Los Baños
September 2021 - July 2026

ORGANIZATIONAL EXPERIENCE

UPLB Ugnayan ng Pahinungód

Co-Facilitator, Maahas Elementary School READ Session | March 2025
- Prepared learning materials suitable for Grade 4-6 students
- Facilitated peer-to-peer reading sessions

Training Participant, Training on Mental Health and Psychosocial Support (MHPSS) | March 2025
- Completed training on handling MHPSS-related situations in a classroom setting

Co-Facilitator, HUME 200a Teambuilding Activity | October 2024
- Facilitated a team-building activity for undergraduate students from the Department of Human and Family Development Studies to enhance their interpersonal skills, teamwork, and communication

Co-Facilitator, Batch 41A, 41B and 42 Volunteer Training Camp | October 2023 - March 2024
- Co-facilitated training sessions for 149 new volunteer

Co-Facilitator, CAS and CEAT NSTP Teambuilding Activity | October 2023
- Facilitated team-building activities for 6 NSTP sections of the College of Arts and Sciences and College of Engineering and Agro-industrial Technology to enhance their group dynamics and equip them with facilitation skills for their literacy projects

UPLB Street Jazz Dance Company

Technical Director | October 2023 - May 2024
- Led a technical committee responsible for stage setup, audio, and lighting
- Coordinated with partner organization to align production plans
- Managed budgeting plans for the committee, including canvassing suppliers, comparing quotations, and negotiating costs

Kapatirang Pitong Lawa sa UP Los Baños

Executive Member | January - April 2023
- Led socio-cultural activities of the organizations
- Managed logistical and technical support for on-ground community activities
- Coordinated with organization alumni and partner organizations for activities
- Coordinated with alumni and event organizers for internal and external events

UPLB College Of Arts and Sciences Freshman Council

External Affairs Committee Co-Head | November 2021 - October 2022
- Handled incoming and outgoing partnership requests from university organizations
- Assisted in overseeing the internal dynamics of the committee

Project Co-Head, CASinag: A Discussion on Student Leadership | March 2022
- Managed the external affairs of the event

- Monitored the benefits and responsibilities of partner organizations
- Coordinated with guest speakers to give information and instructions
- Monitored the development of publicity materials for the event

TRAININGS & CERTIFICATIONS

End-to-End Testing with Playwright | Microsoft
AWS Cloud Practitioner Essentials | Amazon Web Services
Introduction to Cybersecurity | Cisco
Foundations of Cybersecurity | Google